VULNERABILITY ASSESSMENT REPORT

Fellowship Program: Future Interns Cyber Security

Project Code: FUTURE_CS_01

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Target Host: scanme.nmap.org

1. EXECUTIVE SUMMARY

This assessment identifies the security posture of the public-facing host `scanme.nmap.org`. The scan successfully mapped the attack surface, revealing five active services. The overall risk is rated as **Medium**. The primary concern is the presence of legacy software versions for SSH and HTTP services, which are known to have vulnerabilities that have been addressed in more recent releases. Immediate patching is recommended to reduce the risk of unauthorized access or service disruption.

2. ASSESSMENT METHODOLOGY

The scan was conducted using the **Zenmap** graphical interface with the **Nmap 7.99** engine.

- **Scan Parameters:** An "Intense Scan" was executed using the command `nmap -T4 -A -v`.
- **Techniques:** The process included TCP port scanning, service version detection, OS fingerprinting, and the execution of 158 Nmap Scripting Engine (NSE) scripts to identify potential misconfigurations.

3. DETAILED FINDINGS & RISK ANALYSIS

Port 22/TCP | Service: SSH | Version: OpenSSH 6.6.1p1

- **Risk Level:** Medium
- **Analysis:** This version of OpenSSH is associated with an older Ubuntu distribution (2.4.7). Older versions may be susceptible to security weaknesses such as side-channel attacks.
- **Recommendation:** Upgrade to the latest stable OpenSSH version. Implement key-based authentication and disable root login to harden the service.

Port 80/TCP | Service: HTTP | Version: Apache httpd 2.4.7

- **Risk Level:** Medium
- **Analysis:** The web server is running a legacy version of Apache. This version lacks modern security headers and may contain unpatched vulnerabilities related to request smuggling or denial of service.
- **Recommendation:** Update Apache to the current stable branch. Configure the server to hide version strings (ServerTokens Prod) to prevent information leakage.

Port 21/TCP | Service: FTP | Version: tcpwrapped

- **Risk Level:** Low
- **Analysis:** The service responded but provided no detailed version information, suggesting a firewall or wrapper is protecting the port.
- **Recommendation:** Continue monitoring for unauthorized connection attempts and ensure that if FTP is used, it is encrypted via FTPS.

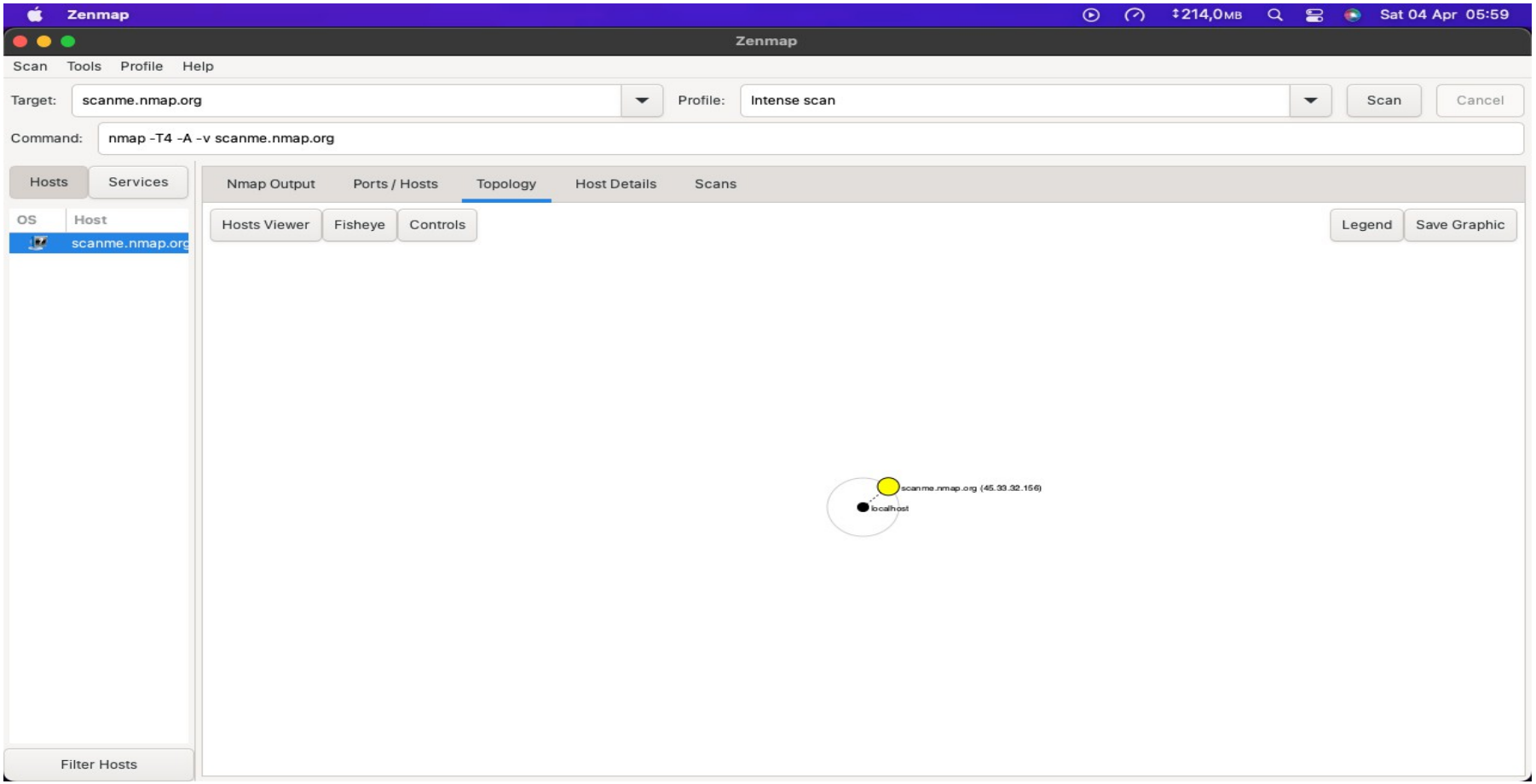
Port 9929/TCP | Service: Nping-echo | Version: Nping echo

- **Risk Level:** Informational
- **Analysis:** This is an echo service used by the Nping tool for network testing. It does not present an immediate vulnerability but identifies the host's utility.
- **Recommendation:** No immediate action required; ensure the service is only reachable by authorized administrators.

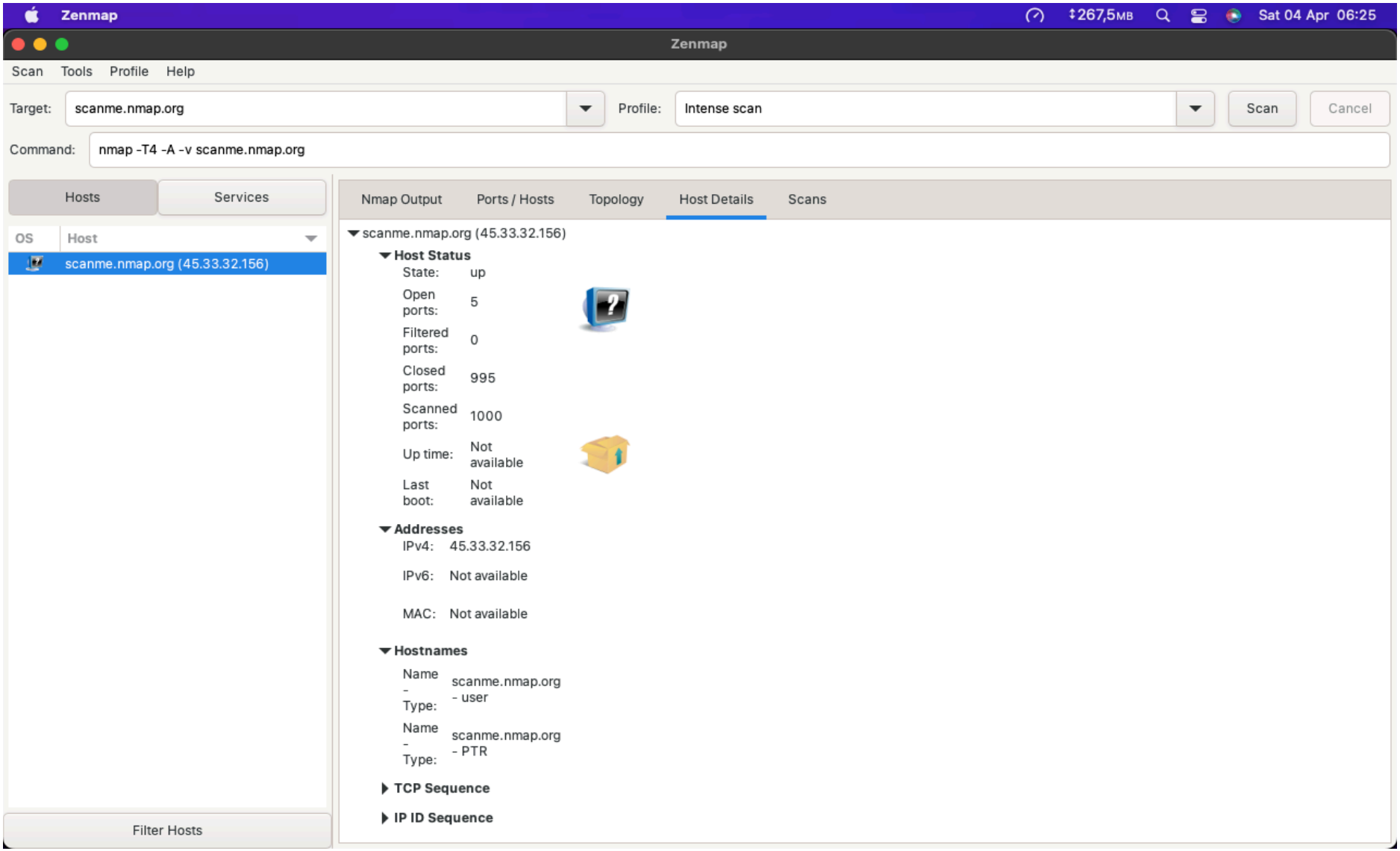
Port 31337/TCP | Service: Elite | Version: tcpwrapped

- **Risk Level:** Low
- **Analysis:** Similar to the FTP port, this service is wrapped. Historically, this port was used by backdoors, but here it appears to be a standard part of the test environment.
- **Recommendation:** Standardize the use of non-standard ports and document their necessity in the business environment.

4. VISUAL EVIDENCE



This diagram provides a visual representation of the network hops required to reach the target host, illustrating the network perimeter.



This confirms the operating system detection and summarizes the status of the 1000 ports scanned during the session.

5. CONCLUSION

The assessment of `scanme.nmap.org` indicates a stable but aging environment. By addressing the outdated software on ports 22 and 80, the administrator can significantly lower the probability of a successful exploit. These findings should be integrated into a regular patch management cycle.